HealthPaddie: Bridging the Health Information Gap

HealthPaddie is an AI-powered health assistant designed to deliver trusted, accessible, and easy-to-understand health information, especially for underserved and rural communities in Africa.

Made with GAMMA

Our Mission: Combating Misinformation

Underserved Communities

Reaching those with limited access to reliable health resources.

Fighting Misinformation

Providing verified health guidance to counter unproven claims.

Accessible Knowledge

Making complex health information simple and understandable.

Made with GAMMA

The Challenge: Barriers to Health Information

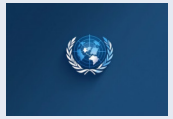
Many individuals rely on unverified sources due to significant barriers in accessing professional medical advice.

- **Limited Literacy:** Health materials are often too complex for those with lower literacy levels.
- **Language Diversity:** Official health information isn't always available in local languages.
- **Access to Professionals:** Scarcity of medical professionals in rural areas creates knowledge gaps.

Made with GAMMA

Building a Foundation of Trust: Verified Data Sources

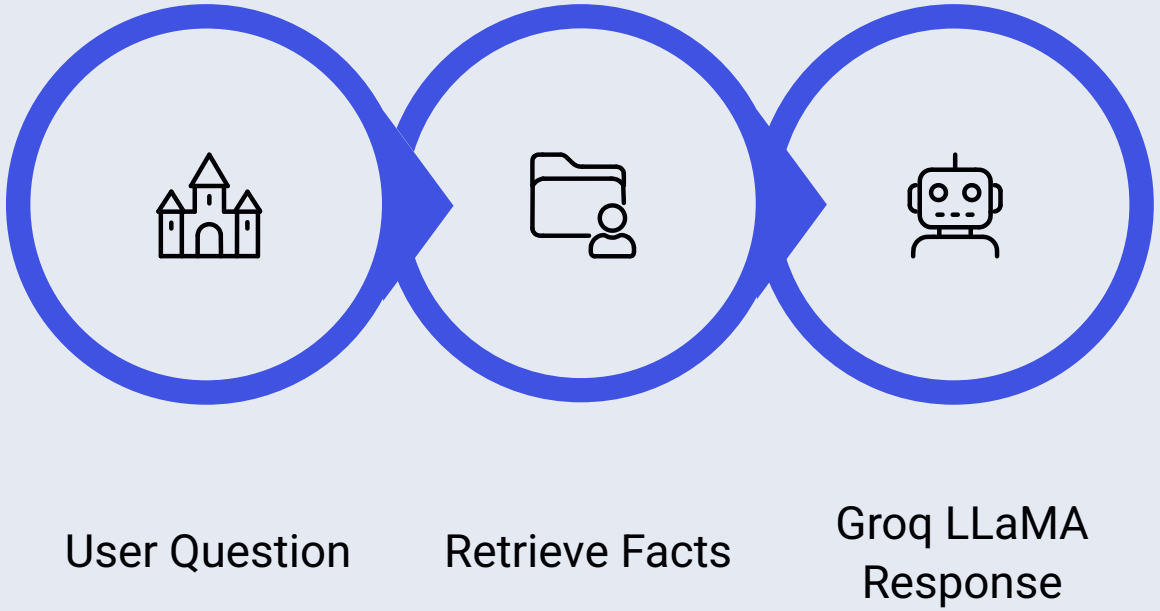
To ensure the highest accuracy, HealthPaddie's knowledge base is built upon credible, verified health documents.



These documents are processed into numerical embeddings using a lightweight sentence transformer model and stored in a FAISS vector database for fast semantic retrieval.

Made with GAMMA

How it Works: Retrieval Augmented Generation (RAG)



Our RAG pipeline minimizes model hallucinations. When a user asks a question, HealthPaddie retrieves the most relevant health facts, inserts them into a structured prompt, and uses a Groq-powered LLaMA 3.1 model to generate a factual, grounded, and safe response.

Made with GAMMA

Empowering Through Language: Multilingual Support

- English
- Hausa
- Yoruba
- Igbo

HealthPaddie supports four major Nigerian languages, allowing users to interact in a natural and familiar way. Our prompt templates ensure clear, culturally appropriate explanations for each language.

Made with GAMMA

Enhanced Accessibility: Text-to-Speech

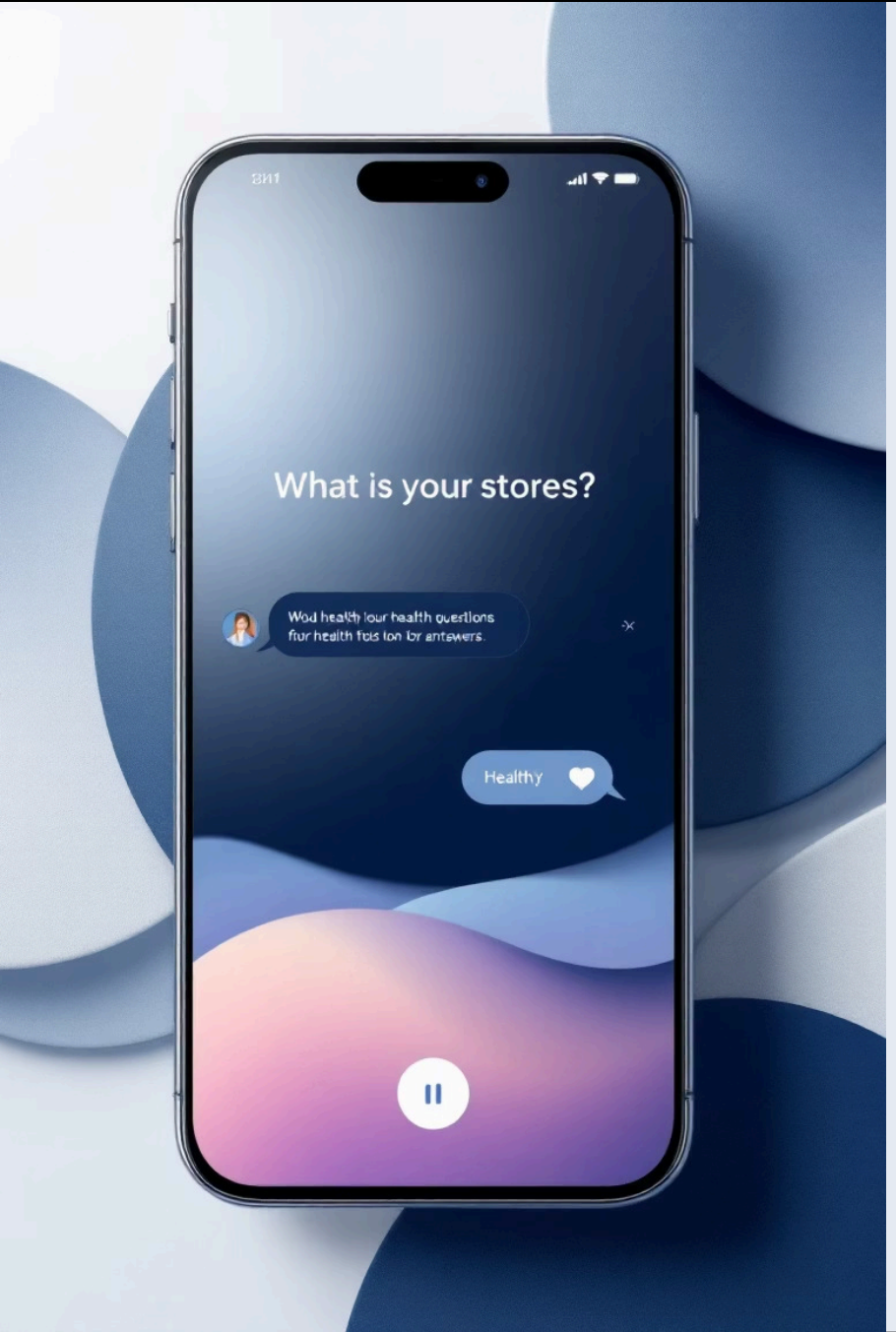
Reaching Every User

For individuals with limited reading ability or visual challenges, HealthPaddie offers an optional text-to-speech output. This feature reads responses aloud, significantly expanding the tool's reach and ensuring everyone can access vital health information.

Inclusive Design

Ensuring information is accessible to all.

Made with GAMMA



User-Friendly Interface: Streamlit Chat

The HealthPaddie interface, built with Streamlit, provides a friendly, chat-style experience. Users can:

- **Select Language:** Choose their preferred language.
- **Ask Questions:** Easily type health inquiries.
- **View History:** Review previous messages.
- **Listen to Answers:** Opt for audio responses.

Made with GAMMA

Responsible AI: Safety First

> "HealthPaddie does not provide medical diagnoses. It warns users when symptoms may indicate a serious condition and encourages them to seek professional care."

Our development prioritizes safe and responsible AI practices. The system prompt enforces strict rules to ensure user safety remains the top priority, guiding users towards professional medical help when necessary.

Made with GAMMA

HealthPaddie: Empowering Health, Together

HealthPaddie demonstrates the power of AI to address critical public health challenges responsibly. By combining trustworthy data, multilingual support, and a user-friendly interface, we empower communities with clear and reliable health information.

Join us in building a healthier future for all.

Made with GAMMA